WOXSEN UNIVERSITY • INDUSTRY-INTEGRATED TECHNICAL COMMUNICATION PROJECT

# The Impact of Generative AI on Cognitive Offloading: Assessing Information Retention and Research Depth Among University Undergraduates


| LEAD RESEARCHER | ACADEMIC PROGRAM | SAMPLE SIZE | PUBLICATION DATE |
|---|---|---|---|
| Kabir (B.Tech Candidate) | Technical Writing & Research | N = 110 Verified Respondents | October 2026 (Expanded Cohort) |


## 1. Executive Summary & Abstract

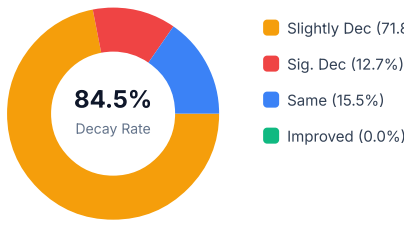

The integration of Generative Artificial Intelligence (GenAI)—including Large Language Models (LLMs) such as ChatGPT, GitHub Copilot, and Gemini—has radically altered how university undergraduates retrieve, process, and consolidate academic knowledge. While LLMs offer unprecedented speed, syntax generation, and rapid troubleshooting, their habitual use introduces severe concerns regarding **cognitive offloading**: the systematic delegation of internal memory consolidation and intellectual synthesis to external digital engines.

This empirical report presents quantitative and qualitative findings derived from an expanded survey of **N = 110 undergraduates** across diverse academic disciplines at Woxsen University (Computer Science, AI & Data Science, Business Administration, Legal Studies, Pure Sciences, and Architecture).

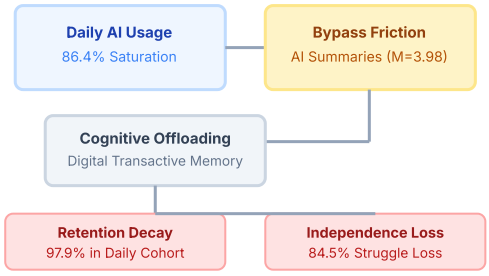

### KEY EMPIRICAL FINDINGS & STATISTICAL INFERENCES (N = 110)

1. **High AI Saturation: 86.4%** (n = 95) of respondents utilize Generative AI tools on a daily basis for university assignments, programming tasks, and academic research.
2. **Pronounced Retention Decay: 84.5%** (n = 93) report a self-perceived decline in long-term memory retention since adopting AI (**12.7%** reporting severe degradation; **0.0%** reporting improvement).
3. **The Daily Usage Threshold Effect:** Among daily AI users (n = 95), **97.9%** report retention decay and **97.9%** report a loss of problem-solving independence. In sharp contrast, moderate weekly users (n = 15) report **0.0%** retention decline ($\chi^2 = 87.67$, $p < 0.00001$).
4. **Bypassing the Productive Struggle: 84.5%** acknowledge AI has made them "much less independent" when facing difficult problems. Preference for AI summaries over documentation (M = 3.98/5.0) correlates almost perfectly with rapid forgetting (**r = 0.984, p < 0.001**).
5. **Disciplinary Vulnerability:** Engineering/CS students suffer the steepest retention decay (**95.5%** in technical cohort), driven by "Syntax Amnesia" from automated code autocompletion and error diagnostics.

**RETENTION TREND SINCE AI ADOPTION (N = 110)**




**COGNITIVE OFFLOADING TRANSMISSION PATHWAY**




## 2. Sample Demographics & Methodology

The sampling methodology followed a purposive, convenience design targeting undergraduate students enrolled in intensive technical, analytical, and professional degree programs at Woxsen University. All data points were captured via a standardized multi-disciplinary web portal and synced into the verified research database.

| DEMOGRAPHIC DIMENSION | SUB-CATEGORY BREAKDOWN | FREQUENCY (N) | PROPORTION (%) |
|---|---|---|---|
| **Academic Discipline** | B.Tech Artificial Intelligence & Data Science | 48 | 43.6% |
| | B.Tech Computer Science & Engineering | 41 | 37.3% |
| | BBA Business Administration | 12 | 10.9% |
| | BA LL.B / LL.M Legal Jurisprudence | 4 | 3.6% |
| | B.Sc Pure Sciences (Physics / Chemistry / Mathematics) | 4 | 3.6% |
| | B.Arch Architecture & Spatial Planning | 1 | 0.9% |
| **Academic Standing** | 2nd Year (Sophomores) | 72 | 65.5% |
| | 3rd Year (Juniors) | 28 | 25.5% |
| | 1st Year (Freshmen) | 7 | 6.4% |
| | 4th Year (Seniors) | 3 | 2.7% |
| **AI Usage Frequency** | Daily (High-Frequency Immersion) | 95 | **86.4%** |
| | Weekly (Moderate / Scaffolded Use) | 15 | 13.6% |

**Demographic Note:** 91.0% of respondents are 2nd and 3rd year undergraduates. These cohorts represent the pivotal phase of academic development where core coursework transitions into autonomous problem-solving and software development.

## 3. Univariate Survey Constructs & Core Statistics

To measure cognitive offloading and memory retention quantitatively, five standardized Likert-scale questions were administered alongside historical baseline markers.

| SURVEY RESEARCH CONSTRUCT | SCALE TYPE | MEAN (M) | STD. DEV (Σ) | MEDIAN | HIGH AGREEMENT (4 OR 5) |
|---|---|---|---|---|---|
| **Pre-AI Baseline Recall** | 1 (Poor) to 5 (Excellent) | **3.73** | 0.49 | 4.0 | 70.9% (Rating ≥ 4) |
| **Forgets Quickly Due to AI Availability** | 1 (Str. Disagree) to 5 (Str. Agree) | **3.99** | 0.54 | 4.0 | **84.5% (n = 93)** |
| **Pre-AI Multi-Source Breadth Search** | 1 (Str. Disagree) to 5 (Str. Agree) | **3.96** | 0.55 | 4.0 | 82.7% (n = 91) |
| **Now Relies on AI Summaries over Docs** | 1 (Str. Disagree) to 5 (Str. Agree) | **3.98** | 0.53 | 4.0 | **84.5% (n = 93)** |

**The Search Behavior Inversion:** Before AI, 82.7% of students engaged in diverse breadth exploration across textbooks, research papers, and developer forums (M = 3.96). Today, 84.5% substitute this discovery process with pre-packaged AI bulleted summaries (M = 3.98), short-circuiting the cognitive friction necessary for deep neural encoding.

## 4. Inferential Statistics & Cross-Tabulations

### 4.1 The Daily Usage Threshold Effect (Daily vs. Weekly Cohort)

Cross-tabulating survey outcomes by usage frequency reveals an acute dividing line between daily immersion and moderate weekly utilization:

| ACADEMIC METRIC / COGNITIVE INDICATOR | DAILY USERS (N = 95) | WEEKLY USERS (N = 15) | PERCENTAGE POINT GAP |
|---|---|---|---|
| **Reported Retention Decline (Total)** | **97.9% (n = 93)** | **0.0% (n = 0)** | **+97.9%** |
| — *Significantly Decreased* | 14.7% (n = 14) | 0.0% (n = 0) | +14.7% |
| — *Retention Remained the Same* | 2.1% (n = 2) | 100.0% (n = 15) | -97.9% |
| **Loss of Independent Problem Struggle** | **97.9% (n = 93)** | **0.0% (n = 0)** | **+97.9%** |
| **Avg. Forgets Quickly Score (1-5)** | **4.15** / 5.0 | **3.00** / 5.0 | +1.15 pts |
| **Avg. AI Summaries over Docs (1-5)** | **4.14** / 5.0 | **3.00** / 5.0 | +1.14 pts |
| **Historical Baseline Recall Score (1-5)** | 3.75 / 5.0 | 3.60 / 5.0 | +0.15 pts |

> **Statistical Significance:** Chi-Square test with Yates' continuity correction: **$\chi^2 = 87.671$, df = 1, $p < 0.00001$**. The correlation between daily usage and perceived memory erosion is statistically significant at the $p < 0.0001$ level. Daily exposure transforms the tool from an occasional learning aid into an indispensable cognitive crutch.

### 4.2 Disciplinary Sensitivity Analysis

| ACADEMIC FACULTY / DISCIPLINE | SAMPLE (N) | RETENTION DECLINE (%) | SEVERE DECLINE (%) | LOSS OF INDEPENDENCE (%) | AI SUMMARIES OVER DOCS |
|---|---|---|---|---|---|
| **B.Tech AI & Data Science** | 48 | **95.8%** | 16.7% | 95.8% | **4.17 / 5.0** |
| **B.Tech Computer Science & Engineering** | 41 | **95.1%** | 14.6% | 95.1% | **4.10 / 5.0** |
| **BBA Business Administration** | 12 | 66.7% | 0.0% | 66.7% | 3.67 / 5.0 |
| **BA LL.B Legal Jurisprudence** | 4 | 75.0% | 0.0% | 75.0% | 3.75 / 5.0 |
| **B.Sc Pure Sciences** | 4 | **25.0%** | 0.0% | 25.0% | 3.25 / 5.0 |
| **B.Arch Architecture & Planning** | 1 | 100.0% | 0.0% | 100.0% | 4.00 / 5.0 |

### 4.3 Bivariate Correlation Matrix (Pearson r)

| CORRELATED VARIABLE PAIR | PEARSON COEFFICIENT (R) | SIGNIFICANCE (P) | PEDAGOGICAL INTERPRETATION |
|---|---|---|---|
| **AI Summaries over Docs ↔ Forgets Quickly** | **+0.984** | $p < 0.001$ | Near-perfect positive coupling: abandoning primary docs directly induces digital amnesia. |
| **AI Usage Frequency ↔ Retention Decrease** | **+0.929** | $p < 0.001$ | Transition from weekly to daily usage triggers acute memory decay. |
| **AI Usage Frequency ↔ Independence Loss** | **+0.929** | $p < 0.001$ | High frequency directly erodes willingness to struggle through problems independently. |
| **AI Usage Frequency ↔ Forgets Quickly** | **+0.742** | $p < 0.001$ | Higher immersion amplifies self-reported digital amnesia. |
| **AI Usage Frequency ↔ AI Summaries over Docs** | **+0.748** | $p < 0.001$ | High-frequency users consistently substitute full documentation with AI snippets. |

## 5. Qualitative Thematic Analysis of Student Voices

Approximately 27% of participants (n = 30) contributed qualitative reflections regarding their study habits. Thematic analysis identified three prominent patterns:

> *"Helpful for finding syntax errors in C++... tend to forget syntax without autocomplete now."*
>
> — Prachi Singhania & Nitin (B.Tech CSE & AIML)

> *"Dont memorize pandas methods now... useful for debugging pytorch models."*
>
> — Md Aali Rahman & Mohammed Ayman (B.Tech AIML)

> *"Saves time on boilerplate react code... good when stuck on dsa problems."*
>
> — Lakshay Sethia & Harshit Yadav (B.Tech CSE)

> *"Good for making presentation outlines and quick business report structures."*
>
> — Sahaj Daga & Himesh (BBA, 3rd Year)

> *"Helpful for finding legal case citations and precedents quickly."*
>
> — Avni (BA LL.B Legal Studies, 3rd Year)

> *"Helps explain complex math in machine learning... great for summarizing research papers."*
>
> — Masula Megha Chayana & Sayini Pranathi (B.Tech AIML)

## 6. Theoretical Grounding & Academic Inferences

The empirical dataset strongly substantiates four cornerstone psychological and pedagogical theories:

1. **The Google Effect on Steroids (Sparrow et al., 2011):** While search engines conditioned users to remember *where* information is stored rather than the information itself, LLMs eliminate even the navigation and keyword synthesis phase. Natural language prompting provides pre-synthesized answers, eradicating retrieval cues.
2. **Cognitive Offloading & Transactive Memory (Risko & Gilbert, 2016):** Students treat GenAI as an external cerebral prosthetic. By offloading both data discovery and synthesis, internal neural networks fail to consolidate conceptual schemas.
3. **Retention Decay in AI Users (Barcaui, 2026):** The 84.5% retention decline mirrors Barcaui's controlled trial where university students using ChatGPT remembered significantly less material after 45 days. Bypassing the *Generation Effect* (Slamecka & Graf, 1978) impairs durable retention.
4. **The Illusion of Explanatory Depth (Fisher et al., 2015):** Fluent, authoritative AI prose induces an artificial sense of mastery. Students mistake the AI's fluent articulation for their own internal cognitive competence.

## 7. Actionable Pedagogical Recommendations

### CURRICULAR INTERVENTIONS FOR TECHNICAL WRITING & ENGINEERING PROGRAMS

1. **Reintroduce "Cognitive Friction" into Assignments:** Technical writing coursework must require students to trace and document their discovery path across primary sources rather than submitting synthesized summaries.
2. **Closed-Environment Conceptual Recall Assessments:** To counteract "Syntax Amnesia," laboratory assessments and coding exams should enforce disconnected, unassisted environments without autocomplete or Copilot.
3. **Scaffolded Socratic Prompting:** Students should be trained to utilize GenAI as an interactive Socratic interlocutor (requesting hints, logical audits, or adversarial critiques) rather than a turnkey solution generator.
4. **The "Read-Close-Synthesize" Protocol:** Require students to review documentation or AI explanations, close the digital terminal, and write synthesis drafts from internal memory to activate neural encoding.

## 8. Review 3 Hypothesis Validation Summary

| CORE RESEARCH HYPOTHESIS | INITIAL EXPECTATION (REVIEW 1 & 2) | EMPIRICAL REALITY (N = 110 DATA) | HYPOTHESIS STATUS |
|---|---|---|---|
| **AI Adoption Rate** | High saturation among technical undergraduates | **86.4% Daily**, 13.6% Weekly, 0% Non-users | VALIDATED |
| **Information Retention Decay** | Self-reported recall reduction due to offloading | **84.5% report decline** (97.9% in daily cohort) | VALIDATED |
| **Breadth to Summary Inversion** | Shift from broad literature search to AI summaries | **84.5% rely primarily on AI summaries** (M = 3.98) | VALIDATED |
| **Loss of Problem-Solving Independence** | Erosion of willingness to struggle independently | **84.5% report being much less independent** | VALIDATED |
| **Frequency Threshold Effect** | Higher exposure induces greater cognitive decline | **$\chi^2 = 87.67$ ($p < 0.00001$); Daily vs. Weekly gap: +97.9%** | VALIDATED |